Raju Molla

PhD Researcher in Computer Science, Keele University

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Research Interests

Artificial Intelligence for Cybersecurity, Agentic AI, Federated Learning, Explainable AI (XAI), Adaptive Intrusion Detection Systems, Adversarial Machine Learning, Trustworthy and Robust Machine Learning, Edge–Cloud–IoT Security, Concept Drift Adaptation, Privacy-Preserving Machine Learning, Anomaly Detection, Distributed Cyber Defence

Education

Keele University **Nov 2026 – Present**
Doctor of Philosophy (PhD) in Computer Science *Keele, United Kingdom*
School of Computer Science and Mathematics (Full-time)

University of Wales Trinity Saint David **June 2025 – June 2026**
MSc in Cybersecurity and Digital Forensics ***(Distinction)*** *London, United Kingdom*
Dissertation: Hybrid Intrusion Detection System for IoT and Cloud Environments

Research Areas: AI for Cybersecurity, Intrusion Detection Systems, IoT Security, Cloud Security, Explainable AI

Eastern University **Oct 2018 – Feb 2023**
BSc in Computer Science and Engineering ***(GPA: 3.74/4.00)*** *Dhaka, Bangladesh*
Research Topic: Cardiovascular Disease Prediction Using Machine Learning

Relevant Areas: Machine Learning, Data Mining, Software Engineering, Algorithms

Research Experience

- **Hybrid Intrusion Detection System for IoT and Cloud Environments**
  Preprint available on *Research Square*.
  DOI: 10.21203/rs.3.rs-10144004/v1
- **Beyond Permissions: Multi-Layer Monitoring and Forensic Behaviour Graph Analysis for Detecting Hidden and Collusive Chrome Extensions**
  Preprint available on *SSRN*.
  SSRN: https://ssrn.com/abstract=6906046
- **IoT-CloudSec: A Multi-Protocol, Multi-Context Simulation Benchmark for Leakage-Aware IoT Intrusion Detection Evaluation**
  Under revision, *IEEE Access*

Professional Experience

Qtec Solution Limited **May 2024 – June 2025**
Software Engineer *Dhaka, Bangladesh*
- Designed and implemented scalable and secure backend systems for financial and healthcare applications
- Developed distributed systems and APIs with emphasis on data integrity and system reliability
- Collaborated with international teams to deploy systems across multiple regions

Bimafy Limited **Dec 2023 – April 2024**
Software Developer *Dhaka, Bangladesh*
- Developed cross-platform mobile applications with integrated cloud services and OCR functionality
- Implemented secure API integrations and real-time application features

Bangla Institute **March 2023 – July 2023**
Backend Engineer Intern *Remote*
- Built backend services and secure authentication systems using Node.js and Express.js
- Optimized database performance and designed scalable data models

Projects

IoT-CloudSec Research Framework **2025–Present**

Python, Scikit-learn, TensorFlow, SHAP

- Implemented a hybrid intrusion detection framework integrating supervised learning and anomaly-aware sequence modelling
- Developed telemetry processing pipelines for IoT, edge, and cloud security datasets
- Performed explainability analysis using SHAP together with robustness and temporal evaluation experiments

UMIS-v2 Distributed Microfinance System **2024–2025**

Asp.net, Razor Page, SQL Server

- Developed scalable backend services for a distributed multi-country microfinance platform deployed across East Africa
- Designed secure APIs and transaction-processing workflows for financial operations
- Improved reliability and scalability of distributed financial services infrastructure

Secure Healthcare API System **2024**

Node.js, Express.js, MongoDB

- Developed secure RESTful APIs for healthcare data management and protected patient information workflows
- Implemented authentication, authorization, and secure data-access mechanisms
- Focused on API security, privacy protection, and backend reliability

Technical Skills

Cybersecurity: Intrusion Detection Systems (IDS), Network Security, Cloud Security, IoT Security, Browser Security, Threat Detection, Threat Analysis, Digital Forensics, OWASP Top 10
Machine Learning & AI: Random Forest, XGBoost, LightGBM, LSTM, Autoencoders, Classification, Anomaly Detection, Explainable AI (SHAP), Feature Engineering, Data Preprocessing
Programming: Python, JavaScript, TypeScript, C, C++, C#
Frameworks & Libraries: Scikit-learn, TensorFlow, Pandas, Node.js, Express.js, React, Next.js, ASP.NET
Tools & Platforms: Kali Linux, Burp Suite, Nmap, Metasploit, SQLMap, FTK Imager, Git, Docker
Databases: MongoDB, SQL Server, MySQL

Achievements

- Top 3% globally on TryHackMe cybersecurity platform
- Completed 130+ hands-on labs in offensive and defensive security
- ICPC Dhaka Regional 2023 participant
- Champion, Eastern University Programming Contests (37th & 38th)